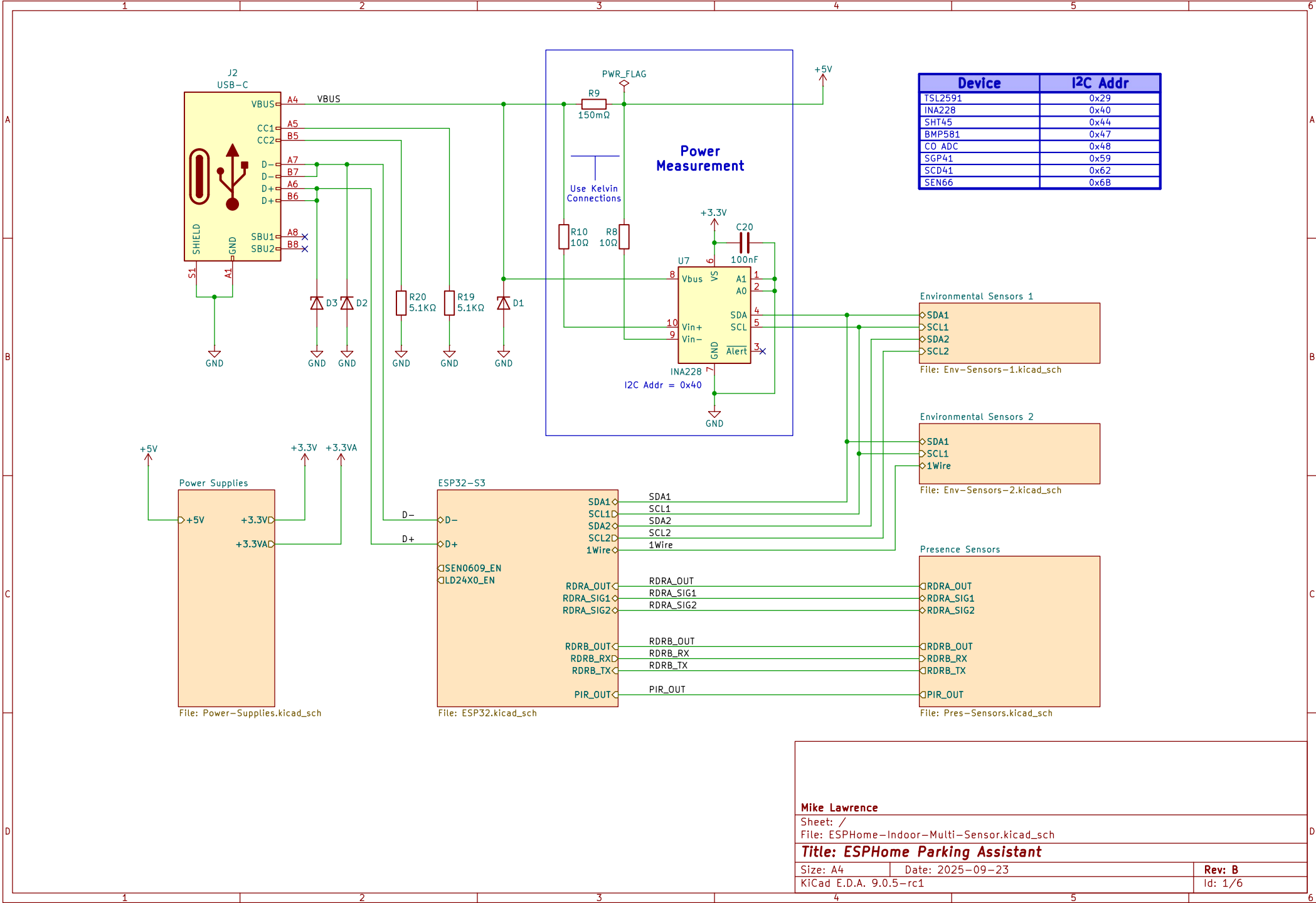




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Sheet: /
File: ESPHome-Indoor-Multi-Sensor.kicad_sch

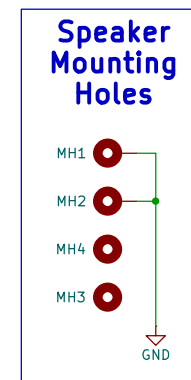
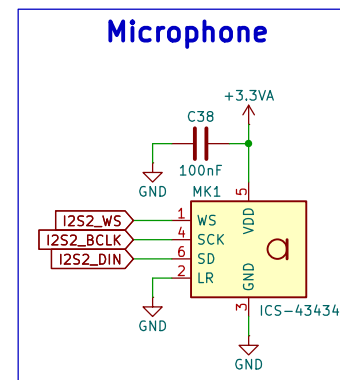
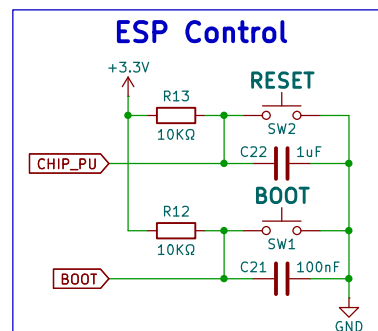
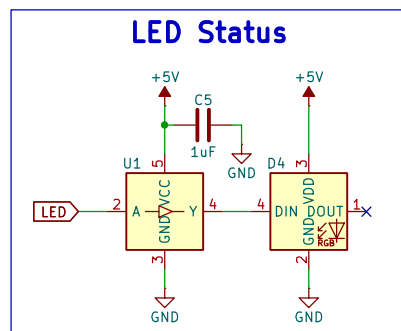
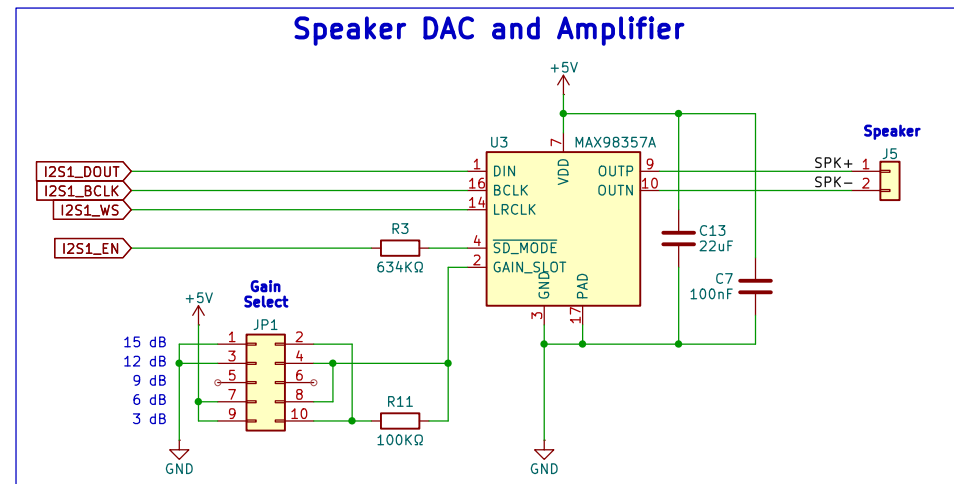
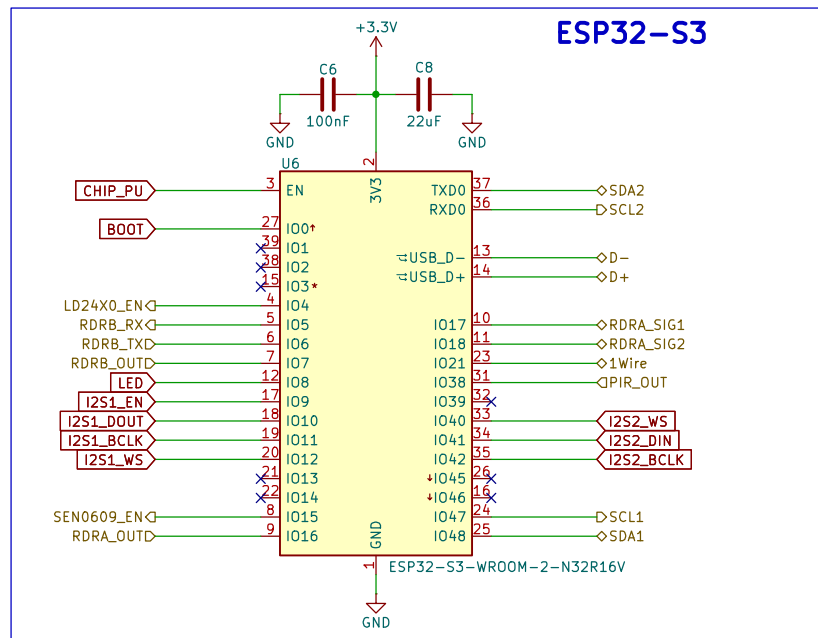
Title: ESPHome Parking Assistant

Size: A4 | Date: 2025-09-23

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Rev: B

Id: 1/6



Notes

1. ESP32-S3 Strapping Pins: GPIO0, GPIO3, GPIO45, GPIO46.
2. MAX98357A DAC when enabled is configured for stereo data mode.
3. SEN0609_EN and LD24X0_EN are no longer used. In order to maintain backwards compatibility these pins should not be used.

Rev B: Support for LD2410S/LD2420 and temp for CO sensor.
Rev A: Fixed Rev - mistakes.

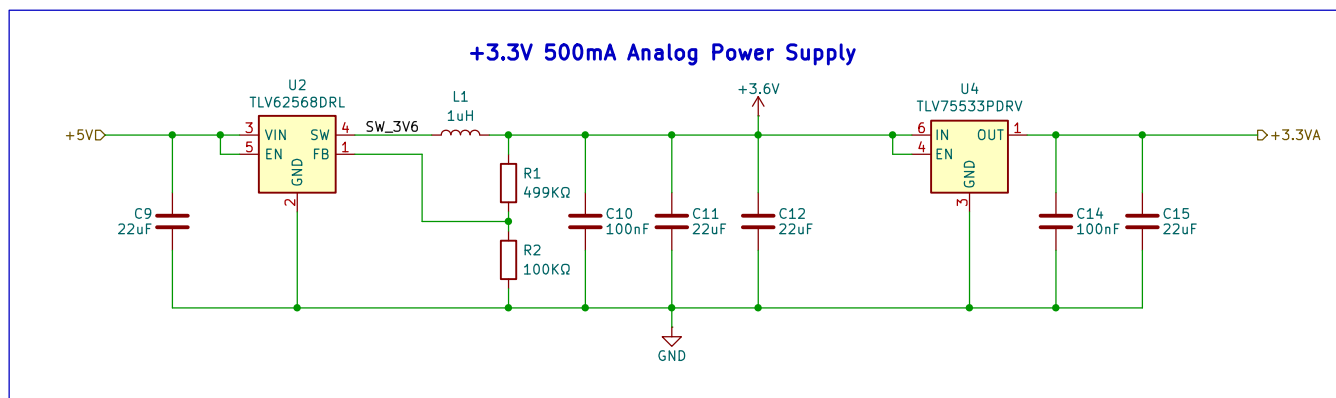
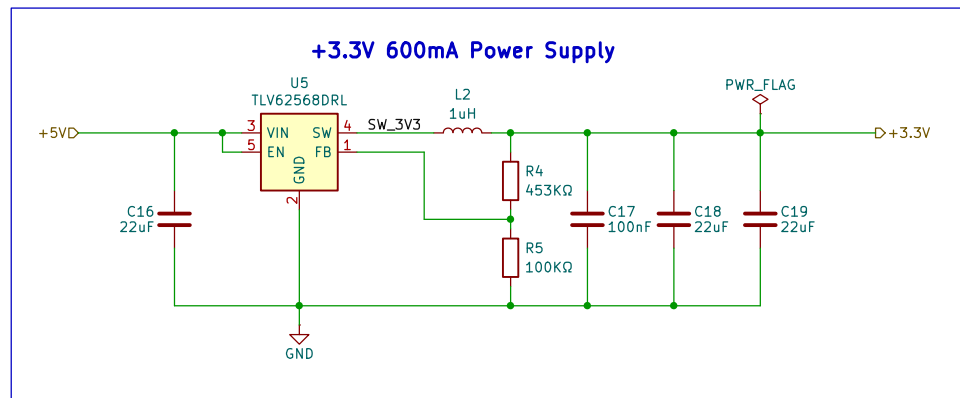
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Sheet: /ESP32-S3/
File: ESP32.kicad_sch

Title: ESPHome Parking Assistant

Size: A4 Date: 2025-09-23
KiCad E.D.A. 9.0.5-rc1

Rev: B
Id: 2/6



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Sheet: /Power Supplies/
File: Power-Supplies.kicad_sch

Title: ESPHome Parking Assistant

Size: A4 Date: 2025-09-23

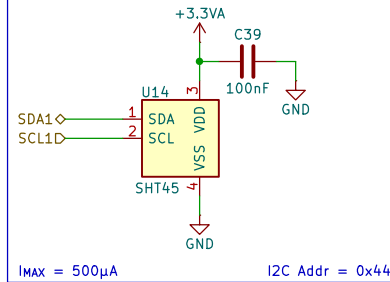
KiCad E.D.A. 9.0.5-rc1

Rev: B

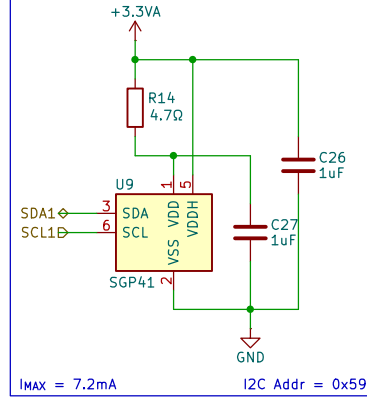
Id: 3/6

Bath or Kitchen Set

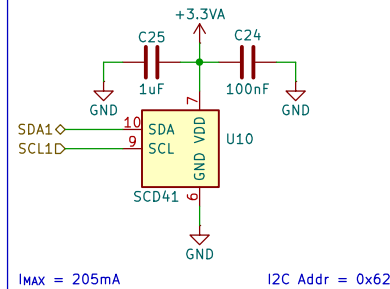
Temperature Humidity Sensor



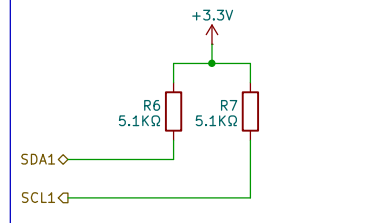
Air Quality Sensor



CO2 Sensor

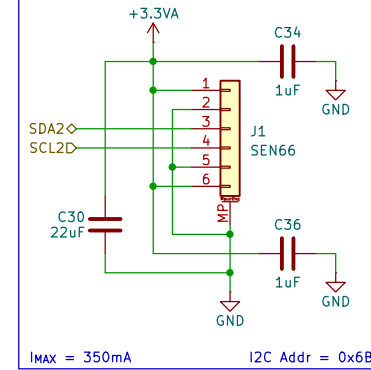


I2C1 Pullup

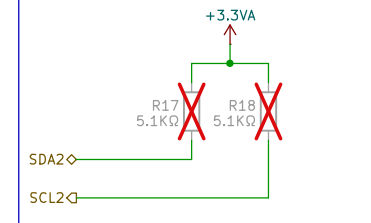


All-In-One Sensor

Environmental Sensor



I2C2 Pullup



Notes

1. Recommend one of two configurations.
 - A. SEN66.
 - B. SHT45, SCD41 and SGP41 for Bath or Kitchen.
2. Temperature & Humidity Sensor can be SHT40, SHT41, or SHT45 for increasing accuracy.
3. CO2 (SCD40 or SCD41) measures CO2, Temperature and Humidity.
4. Air Quality Sensor (SGP40 or SGP41) measures VOC, NOx. These sensors are self heating and not recommended for measuring room temperature.
5. SEN66 measures Temperature, Humidity, CO2, VOC, NOx and PM.

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Sheet: /Environmental Sensors 1/
File: Env-Sensors-1.kicad_sch

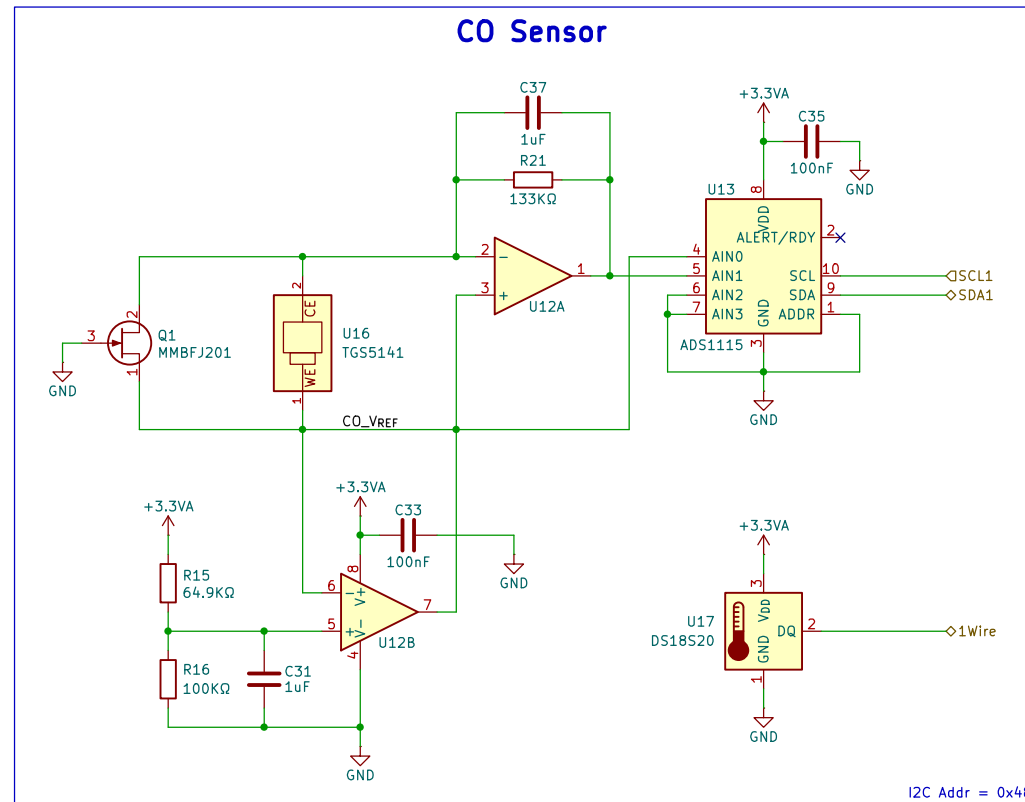
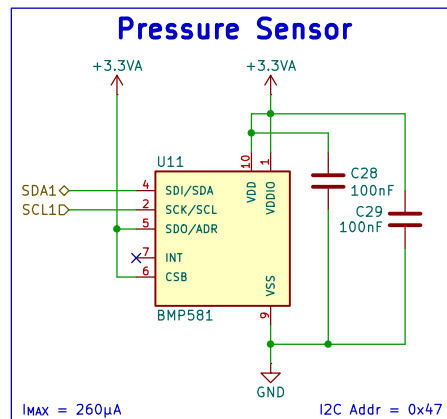
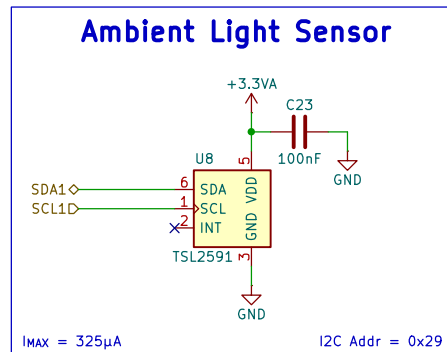
Title: ESPHome Parking Assistant

Size: A4 Date: 2025-09-23

KiCad E.D.A. 9.0.5-rc1

Rev: B

Id: 4/6



Notes

1. Accuracy of CO_VREF not critical.
Needs to be around 2V to turn off Q1.
2. ADC range 0.512V.
3. Max 1000PPM.
4. Gain = $0.512V / (3.2nA \times 1.225 \times 1000PPM) = 133.333k$

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Sheet: /Environmental Sensors 2/

File: Env-Sensors-2.kicad_sch

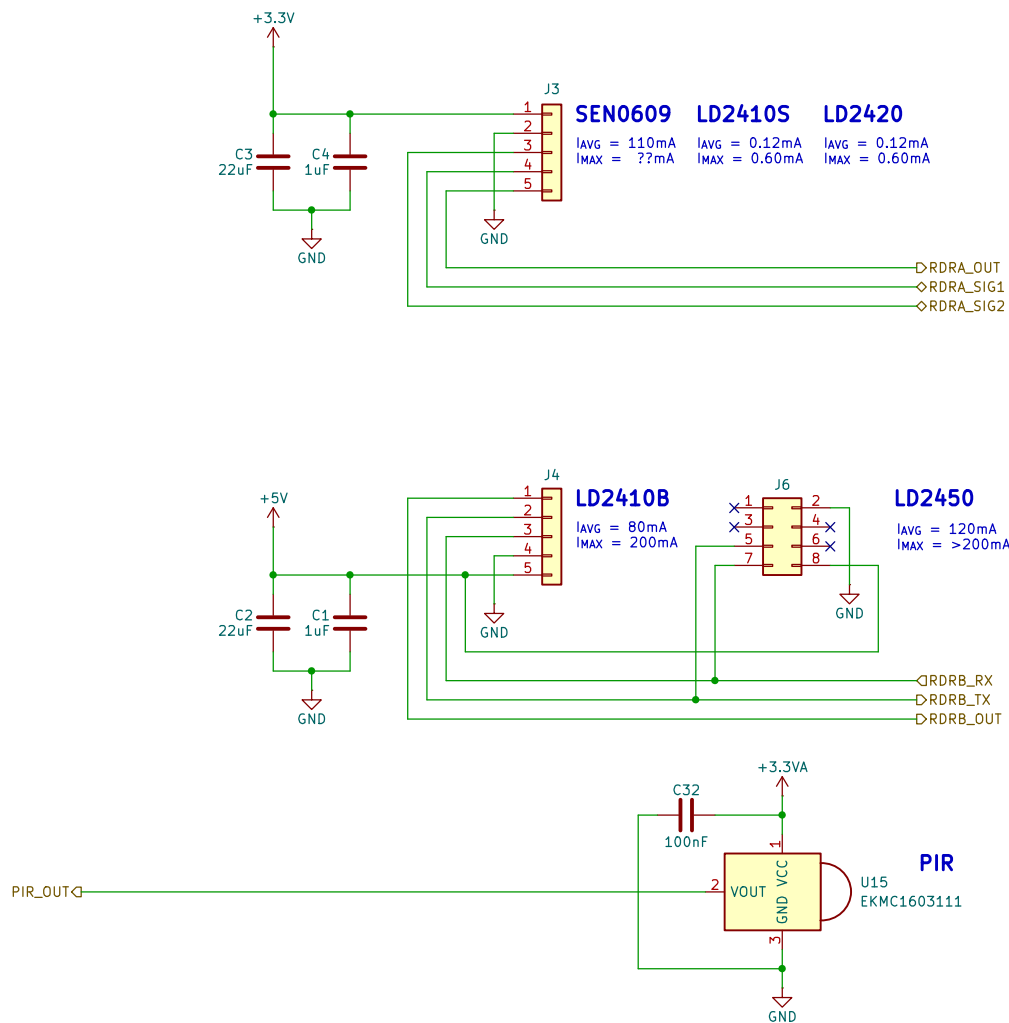
Title: ESPHome Parking Assistant

Size: A4 Date: 2025-09-23

KiCad E.D.A. 9.0.5-rc1

Rev: B

Id: 5/6



Notes

1. RDRA_SIG1 is SEN0609 TX, and LD2410S/LD2420 RX.
2. RDRA_SIG2 is SEN0609 RX, and LD2410S/LD2420 TX.

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Sheet: /Presence Sensors/
File: Pres-Sensors.kicad_sch

Title: ESPHome Parking Assistant

Size: A4 Date: 2025-09-23

KiCad E.D.A. 9.0.5-rc1

Rev: B

Id: 6/6